

Migration Validation & Risk Audit

Client: TestClient
Migration: TestMig
Date: 2026-02-21
Auditor: Independent Migration Audit

Executive Summary

Final Verdict: NO-GO	
Data Volume Checks	PASS
Aggregate Checks	PASS
Mapping Checks	PASS
Data Constraint Checks	PASS
Relationship Checks	FAIL

Data Volume Checks

Check Name	Status	Details
Volume Check: users	PASS	Row counts match exactly for table 'users'. Source: 50, Target: 50 rows.
Volume Check: orders	PASS	Row counts match exactly for table 'orders'. Source: 80, Target: 80 rows.
Volume Check: order_items	PASS	Row counts match exactly for table 'order_items'. Source: 120, Target: 120 rows.

Volume Check: products

PASS

Row counts match exactly for table 'products'. Source: 30, Target: 30 rows.

Aggregate Checks

Check Name	Status	Details
Sum Check: users - age	PASS	Sum matches exactly for column 'age' in table 'users'. Source and Target both have sum 1959. {'pct_difference': np.float64(0.0)}
Average Check: users - age	PASS	Average matches exactly for column 'age' in table 'users'. Source and Target both have average 39.18. {'pct_difference': np.float64(0.0)}
Max Check: users - age	PASS	Max matches exactly for column 'age' in table 'users'. Source and Target both have max 55. {'pct_difference': np.float64(0.0)}
Min Check: users - age	PASS	Min matches exactly for column 'age' in table 'users'. Source and Target both have min 25. {'pct_difference': np.float64(0.0)}
Variance Check: users - age	PASS	Variance matches exactly for column 'age' in table 'users'. Source and Target both have variance 96.885306122449. {'pct_difference': np.float64(0.0)}
Sum Check: users - tenure_days	PASS	Sum matches exactly for column 'tenure_days' in table 'users'. Source and Target both have sum 53368. {'pct_difference': np.float64(0.0)}
Average Check: users - tenure_days	PASS	Average matches exactly for column 'tenure_days' in table 'users'. Source and Target both have average 1067.36. {'pct_difference': np.float64(0.0)}

Max Check: users - tenure_days	PASS	Max matches exactly for column 'tenure_days' in table 'users'. Source and Target both have max 1972. {'pct_difference': np.float64(0.0)}
Min Check: users - tenure_days	PASS	Min matches exactly for column 'tenure_days' in table 'users'. Source and Target both have min 166. {'pct_difference': np.float64(0.0)}
Variance Check: users - tenure_days	PASS	Variance matches exactly for column 'tenure_days' in table 'users'. Source and Target both have variance 309923.37795918365. {'pct_difference': np.float64(0.0)}
Sum Check: orders - amount	PASS	Sum matches exactly for column 'amount' in table 'orders'. Source and Target both have sum 83775.83000000002. {'pct_difference': np.float64(0.0)}
Average Check: orders - amount	PASS	Average matches exactly for column 'amount' in table 'orders'. Source and Target both have average 1047.1978750000003. {'pct_difference': np.float64(0.0)}
Max Check: orders - amount	PASS	Max matches exactly for column 'amount' in table 'orders'. Source and Target both have max 1992.23. {'pct_difference': np.float64(0.0)}
Min Check: orders - amount	PASS	Min matches exactly for column 'amount' in table 'orders'. Source and Target both have min 57.09. {'pct_difference': np.float64(0.0)}
Variance Check: orders - amount	PASS	Variance matches exactly for column 'amount' in table 'orders'. Source and Target both have variance 278405.6154928956. {'pct_difference': np.float64(0.0)}
Sum Check: orders - quantity	PASS	Sum matches exactly for column 'quantity' in table 'orders'. Source and Target both have sum 449. {'pct_difference': np.float64(0.0)}

Average Check: orders - quantity	PASS	Average matches exactly for column 'quantity' in table 'orders'. Source and Target both have average 5.6125. {'pct_difference': np.float64(0.0)}
Max Check: orders - quantity	PASS	Max matches exactly for column 'quantity' in table 'orders'. Source and Target both have max 10. {'pct_difference': np.float64(0.0)}
Min Check: orders - quantity	PASS	Min matches exactly for column 'quantity' in table 'orders'. Source and Target both have min 1. {'pct_difference': np.float64(0.0)}
Variance Check: orders - quantity	PASS	Variance matches exactly for column 'quantity' in table 'orders'. Source and Target both have variance 7.784651898734177. {'pct_difference': np.float64(0.0)}
Sum Check: order_items - price	PASS	Sum matches exactly for column 'price' in table 'order_items'. Source and Target both have sum 28328.28. {'pct_difference': np.float64(0.0)}
Average Check: order_items - price	PASS	Average matches exactly for column 'price' in table 'order_items'. Source and Target both have average 236.069. {'pct_difference': np.float64(0.0)}
Max Check: order_items - price	PASS	Max matches exactly for column 'price' in table 'order_items'. Source and Target both have max 493.89. {'pct_difference': np.float64(0.0)}
Min Check: order_items - price	PASS	Min matches exactly for column 'price' in table 'order_items'. Source and Target both have min 6.32. {'pct_difference': np.float64(0.0)}
Variance Check: order_items - price	PASS	Variance matches exactly for column 'price' in table 'order_items'. Source and Target both have variance 20123.940056134452. {'pct_difference': np.float64(0.0)}

Sum Check: order_items - quantity	PASS	Sum matches exactly for column 'quantity' in table 'order_items'. Source and Target both have sum 361. {'pct_difference': np.float64(0.0)}
Average Check: order_items - quantity	PASS	Average matches exactly for column 'quantity' in table 'order_items'. Source and Target both have average 3.0083333333333333. {'pct_difference': np.float64(0.0)}
Max Check: order_items - quantity	PASS	Max matches exactly for column 'quantity' in table 'order_items'. Source and Target both have max 5. {'pct_difference': np.float64(0.0)}
Min Check: order_items - quantity	PASS	Min matches exactly for column 'quantity' in table 'order_items'. Source and Target both have min 1. {'pct_difference': np.float64(0.0)}
Variance Check: order_items - quantity	PASS	Variance matches exactly for column 'quantity' in table 'order_items'. Source and Target both have variance 1.8570728291316525. {'pct_difference': np.float64(0.0)}
Sum Check: products - price	PASS	Sum matches exactly for column 'price' in table 'products'. Source and Target both have sum 12119.300000000001. {'pct_difference': np.float64(0.0)}
Average Check: products - price	PASS	Average matches exactly for column 'price' in table 'products'. Source and Target both have average 403.9766666666667. {'pct_difference': np.float64(0.0)}
Max Check: products - price	PASS	Max matches exactly for column 'price' in table 'products'. Source and Target both have max 968.18. {'pct_difference': np.float64(0.0)}
Min Check: products - price	PASS	Min matches exactly for column 'price' in table 'products'. Source and Target both have min 55.28. {'pct_difference': np.float64(0.0)}

Variance Check: products - price	PASS	Variance matches exactly for column 'price' in table 'products'. Source and Target both have variance 97809.61317471262. <code>{'pct_difference': np.float64(0.0)}</code>
Sum Check: products - stock_quantity	PASS	Sum matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have sum 7410. <code>{'pct_difference': np.float64(0.0)}</code>
Average Check: products - stock_quantity	PASS	Average matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have average 247.0. <code>{'pct_difference': np.float64(0.0)}</code>
Max Check: products - stock_quantity	PASS	Max matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have max 496. <code>{'pct_difference': np.float64(0.0)}</code>
Min Check: products - stock_quantity	PASS	Min matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have min 17. <code>{'pct_difference': np.float64(0.0)}</code>
Variance Check: products - stock_quantity	PASS	Variance matches exactly for column 'stock_quantity' in table 'products'. Source and Target both have variance 24260.96551724138. <code>{'pct_difference': np.float64(0.0)}</code>

Mapping Checks

Check Name	Status	Details
Mapping Check: users	PASS	All mappings are valid for table 'users'.
Mapping Check: users	PASS	All mappings are valid for table 'users'.
Mapping Check: orders	PASS	All mappings are valid for table 'orders'.

Mapping Check: orders	PASS	All mappings are valid for table 'orders'.
Mapping Check: order_items	PASS	All mappings are valid for table 'order_items'.
Mapping Check: products	PASS	All mappings are valid for table 'products'.
Mapping Check: products	PASS	All mappings are valid for table 'products'.

Data Constraint Checks

Check Name	Status	Details
Data Constraints Check: users.first_name	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.last_name	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.email	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.date_of_birth	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: users.date_onboarded	PASS	All data constraints are satisfied for table 'users'.
Data Constraints Check: orders.order_date	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: orders.amount	PASS	All data constraints are satisfied for table 'orders'.
Data Constraints Check: orders.user_id	PASS	All data constraints are satisfied for table 'orders'.

Data Constraints Check: order_items.order_id	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.product_id	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.quantity	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: order_items.price	PASS	All data constraints are satisfied for table 'order_items'.
Data Constraints Check: products.product_name	PASS	All data constraints are satisfied for table 'products'.
Data Constraints Check: products.price	PASS	All data constraints are satisfied for table 'products'.
Data Constraints Check: products.stock_quantity	PASS	All data constraints are satisfied for table 'products'.

Relationship Checks

Check Name	Status	Details
Foreign Key Check: orders	PASS	All foreign key values in 'user_id' of child table 'orders' have matching primary keys in parent table.
Foreign Key check (Missing FK): orders	FAIL	Foreign key column 'order_id' not found in child table 'orders'.
Foreign Key Check: order_items	PASS	All foreign key values in 'order_id' of child table 'order_items' have matching primary keys in parent table.

Foreign Key Check:
order_items

PASS

All foreign key values in 'product_id' of child table
'order_items' have matching primary keys in parent
table.

Final Deployability Verdict

NO-GO
